

Extremophile Farm



Introduction.

This project investigates the creation and implementation of a simulated environment using the coding development environment Processing and the Java language. It was inspired by the Smithsonian Institute's website detailing the lifecycle of Microbes in the Ocean and investigations of interactions that take place around geothermal vents.

The Problem.

In the Deep Ocean, beyond human intervention there are creatures that have evolved in their own specific ecological niches. This environment is so remote and extreme that survival seems to be an impossibility. How do you inspire people to learn more about this strange and inaccessible place? What happens when they interfere with it?

Approach.

Simulating the complex environment that exists around a deep undersea volcano, by necessity required some simplification. Chemical reactions were grouped into a single temperature variable. Organisms and their interactions were constrained to show basic predator/ prey dynamics and excluded symbiotic relationships.

Concept.

Extremophile Farm is based around a single hydrothermal vent, or the Black Smoker. The user can manipulate the environment and its inhabitants by adding creatures and adjusting variables with sliders that change temperature and the amount of primary nutrient that are available. This nutrient is depicted as Marine Snow (Snow), and represents organic detritus that falls from the upper ocean column. It is consumed by Tube worms that live close to the Black Smoker. At the Ocean Floor, Snow changes into bacteria, where it is "farmed" or eaten by the Hoff Crab. This Crab is predated on by the cautious, but hungry Vulcanoctopus. These inhabitants are represented using emoji characters to create an accessible, yet familiar and engaging simulation in what is in reality a hostile and dangerous environment.

Interface Rationale.

User interface controls allow the environmental manipulation of temperature, nutrient density and seismic events. At the Top-Bar UX environmental inputs using sliders gradual changes can be altered and changes observed directly. The Side-Bar area mimics the water column, and follows familiar conventional layout styles using labeled buttons. A population counter shows living organisms and allows the user to correct an imbalance or prevent an extinction. Continuous feedback with the Bacterial Level indicator provides early warning for a threshold "Bacteria Overpopulation" event telling the user that the system has collapsed. In an active volcanic region, the danger of an eruption is imminent. The user can choose to trigger an eruption by pressing a bright "danger" red button.

Outcome.

From the start the ecosystem is in balance, by design. It is user input that changes the conditions of survival. Too much marine snow can accumulate and cause a bacteria overpopulation, too little will cause the Hoff Crabs to starve. The on flow of this will also cause the Vulcanoctopus to starve. In the ecosystem Tube Worms persist, but can be killed out if the temperature increases or decreases too quickly.

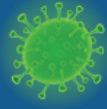
A user triggered series of eruptions can end all life or it can cull an overpopulated species and allow balance to be maintained. These mechanisms show that the ecosystem as it is runs in balance, and yet encourages the user to experiment and explore for themselves.

Ecosystem Layout and Design.

Figure 1.



Figure 2.



Environment

Marine Snow: Figure 1. Marine Snow falls from the top of the screen. A user operated slider increases or decreases the Snow density.

Bacteria, Figure 2. Marine Snow turns into Bacteria when it reaches the Ocean Floor.

Figure 3.



The Black Smoker: Figure 3. Heat Source, heat spreads and size increases with the user controlled slider. This element is a source of destruction if the user selects the eruption option.

Source of Temperature, 453°C at the centre.

Linked to Eruption button, which ejects rocks.

Organisms

Figure 4.



Tube worm: Figure 4.

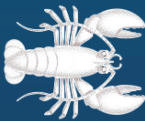
Lives close to the Black Smoker.

Dies if it is outside the temperature range (90°C to 420°C).

Spawns when a colony consumes 2 falling Marine Snow.

Can be spawned by the user.

Figure 5.



Hoff Crabs: Figure 5.

Eats bacteria. Feeding pattern is along the bottom.

Can be spawned by the user.

Spawns if an individual consumes 4 Bacteria.

Prefers 3°C to 30°C

Loses vitality and eventually dies above 30°C.

Can be spawned by the user.

Figure 6.



Vulconoctopus: Figure 6.

Eats Hoff Crabs.

Lives between 300°C and 45°C.

Can be spawned by the user.

Spawns if an individual consumes 4 Hoff Crabs.

Can be spawned by the user.

Ecosystem Layout and Display.

Figure 7. Ecosystem as depicted in running Simulation.



Note. Screenshot of Ecosystem

Figure 8. Interactions Matrix

Black Smoker (Volcano) Temp 3°C - 452°C (Heat radiated from the base)	Marine Snow + Bacteria (Detritus falling from the upper levels becomes bacteria when it settles on the Ocean Floor)	Tube Worm Temp Range 90°C - 452°C	Hoff Crab (Mobile Scavenger)	Vulconoctopus (Mobile Scavenger) Lives close to the Black Smoker
Marine Snow + Bacteria (Detritus falling from the upper levels becomes bacteria when it settles on the Ocean Floor)	No Interaction	Eats Marine Snow as it falls from the top.	No Interaction	No Interaction
Tube Worm Survives in Proximity to Black Smoker. Temp Range 90°C - 452°C	Eats Marine Snow	No Interaction	No Interaction.	Tube worms are Habitat for Vulconoctopus
Hoff Crab Survives in Proximity to Black Smoker. Temp 3°C - 30°C	Eats Accumulated Bacteria	No Interaction	No Interaction	Preferred Food Source is Hoff Crab.
Vulconoctopus (Mobile Scavenger) Lives close to the Black Smoker	No Interaction	Will Die if the bacteria level covers it.	Avoids Vulconoctopus.	No Interaction

Note. Ecosystem relationships reference guide.

UX Layout, Design and Operation.

The user interface area as depicted in the design sketch allows a simple layout that conforms to the "F" pattern.

Scanning across the top, the simulation name is bold and large. Sliders that manipulate Temperature and Snow density are accessible and labeled with their function.

Population is shown in the sidebar. This display has the organism and their count.

An option to add a new creature is given using the universal "+" as a button icon.

Speed control uses a radio button that changes colour to show which speed is active.

The Eruption button in red signals danger or caution.

Pause and Reset are in-line underneath.

The bacteria Level indicator is a bacteria green bar that goes up as bacteria increases, this triggers an end state when the level is too high to sustain the population.

UX Rationale

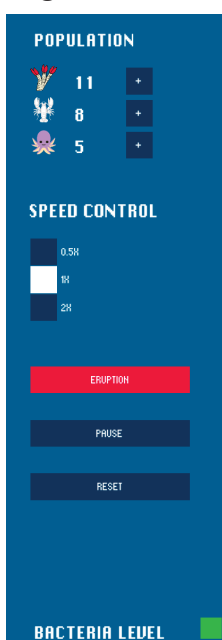
The reason for integrating buttons and sliders is so that the user is supported in their interaction with the use of familiar icons and immediate feedback. These controls encourage experimentation, observation and engagement.

Figure 9. Top Bar User Interface Area



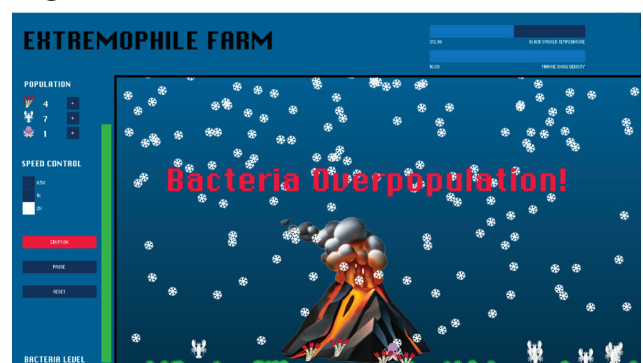
Note. Left, Simulation Name. Right, Sliders for Temperature and Marine Snow changes.

Figure 10. Side Bar User Interface Area



Note. Add Creature Buttons, Population Speed Pause, Reset and Bacteria Level Indicator.

Figure 11. Simulation End State



Note. Bacteria Level is full, the user is informed the simulation has stopped.

References

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Notes on Ai Usage.

I used Claude to create a README reference file to catalog all of the concepts used in the Studio Exercises.

Also used for concept development, planning and feedback to make sure the written report covered and fit the supplied brief.

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Appendix

Extremophile Farm - Mood board Deep Sea Hydrothermal Vent Ecosystem

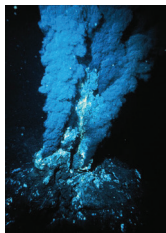
In the Abyssal Zone, this area that covers 60% of the Ocean at depths of 4000 to 6000 Metres under the sea there is an active hydrothermal vent. It's dark, but teeming with life. Burps of sulfur rich mineral deposits stream from a Black Smoker. Tube worms sway in the Ocean Current, Hoff Crabs and Bacteria thrive.

All the while Marine Snow, a nutrient rich detritus falls into this void.

In spite of the many symbiotic relationships, these lifeforms compete to survive.

Our player can manipulate Time, Temperature and Nutrient Density, observing what survives at the extreme edge of nature.

Figure 12.
Black Smoker



Note.
Deep Sea
Hydrothermal Vent

Figure 13.
Tube worms



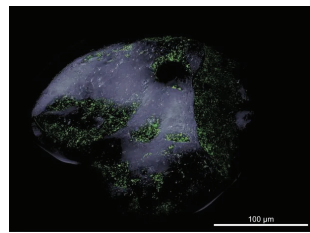
Note.
Tube worms feeding
in a cluster.

Figure 14.
Hoff Crab

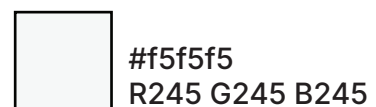
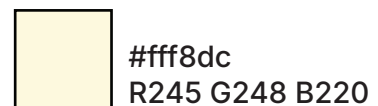
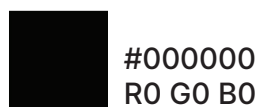
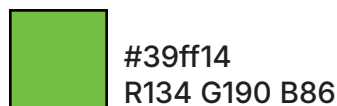
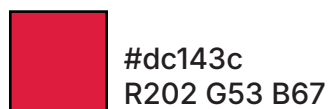
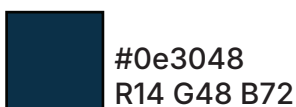
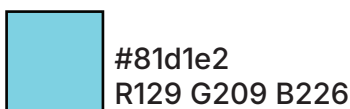


Note.
Bacteria eating lobster
clusters at 700/ Msq.

Figure 15.
Microbes



Note.
Microbial colonization
of a sand grain.



Appendix

User Persona: Sam



Age: 12 years old.

Role: Student / The budding Scientist.

Tech Literacy: High (she grew up with tablets), but prefers intuitive visuals over reading long manuals.

Motivation: Discovery through "breaking" things and seeing cause-and-effect.

Goals & Needs

Visual Engagement: Familiar looking creatures that interact.

Motivations: Inspired by the fact that the deep ocean is less explored than the moon. She wants to feel like an explorer.

Instant Feedback: Needs to see an immediate action when moving a slider.

Tech Behavior: Slowly adjusts the controls and observes behaviour.

Ease of Use: Buttons and sliders should be large, obvious, and clear labels

Goal: To understand how biological balance works in extreme environments. She wants to find the exact "equilibrium".

Pain Points

Frustrations: If everything dies without an understanding as to why balance was lost.

Understanding causal effect: Why did the Bacteria Overpopulate?

Traits

Experimental: Loves pushing variables to their extremes to see what happens.

Visual Learner: Prefers watching the animation.

Impatient: If the "speed" isn't adjustable, they might get bored waiting for reproduction.

Sam's User Journey

Enter: Sees a dark, mysterious underwater Volcano.

Interact: Moves the temperature slider to 100%.

Reaction: Watches the tube worms multiply and watches what happens if the temperature increases.

Outcome: Understands the delicate balance of extremophile life through play.

Note: Original template created using Generative Ai, then refined.

Appendix

User Persona: Jack



Age: 22 years old.

Role: Student

Tech Literacy: High, Studying Design at University of Waikato

Motivation: Interested in Science Communication and Graphic Design.

Goals & Needs

Visual Engagement: An appreciator of good layout and retro aesthetics.

Inspiration: Wants to see interesting projects getting made and uses that to make their own things.

Goal: To test the limits of the simulation. He wants to see what happens when it's not in balance.

Motivations: He enjoys "God-mode" simulations where he can manipulate variables and observe chaotic results.

Tech Behavior: He will Likely use the Speed Control (1 times or faster) to see long-term effects quickly. He'll add maximum Marine Snow to see if the lobsters can keep up.

Pain Points

Boring Visuals: Not used to waiting for things to load, easily loses interest if there is nothing happening.

Manipulation: Wants to be able to adjust and change parameters.

Frustrations: Doesn't want to sit and wait, would rather speed up the simulation.

Traits

Curiosity: Wants to push the simulation And see if they can max out the conditions.

Metrics: Interested in numbers and knowing how many organisms are in the habitat.

Jack's User Journey

Enter: Sees a detailed and engaging presentation.

Interact: Quickly adds a whole lot of organisms.

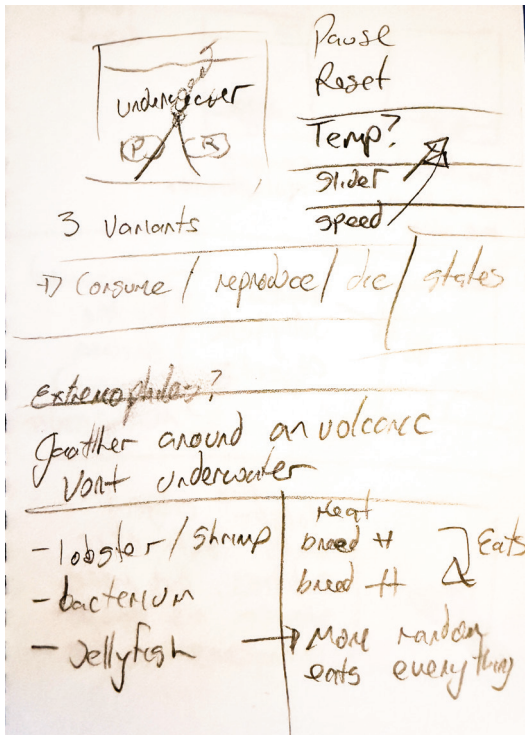
Reaction: Seeing things happens they dial up the speed and Temperature.

Outcome: Pushes reset and tries a different combination.

Note: Original template created using Generative Ai, then refined.

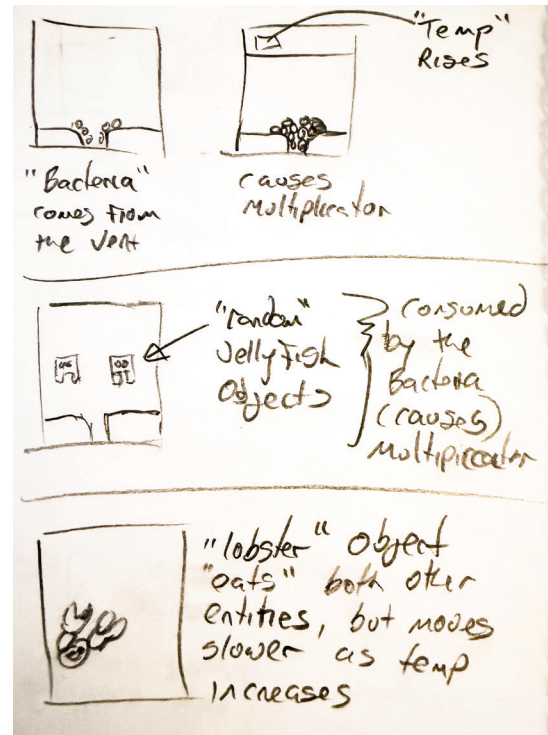
Appendix

Figure 16. Thumbnail Sketch 1



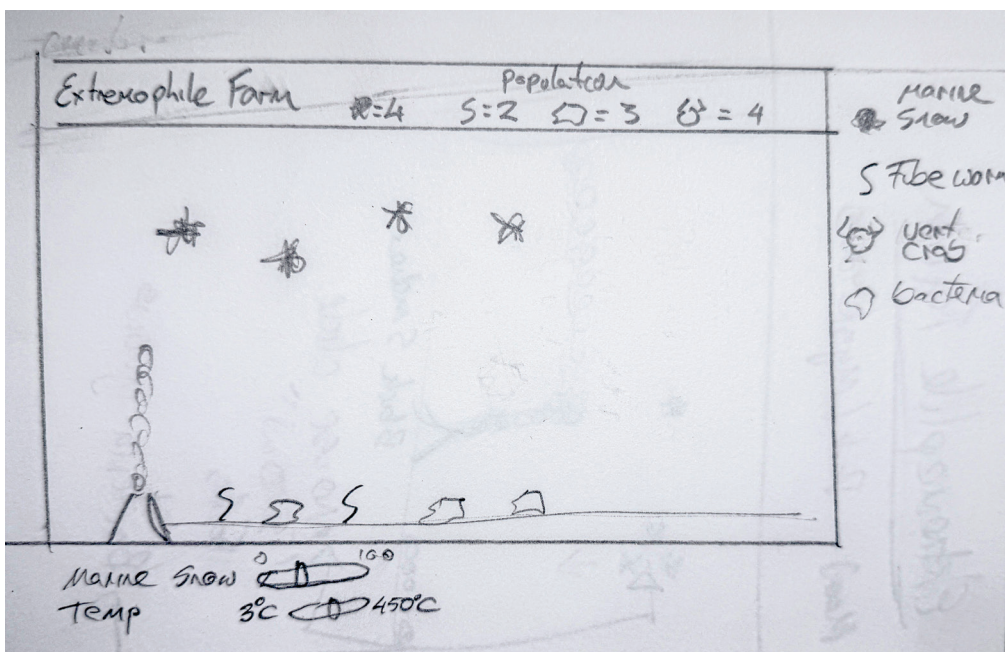
Note. Conceptual Development

Figure 17. Thumbnail Sketch 2



Note. Conceptual Development

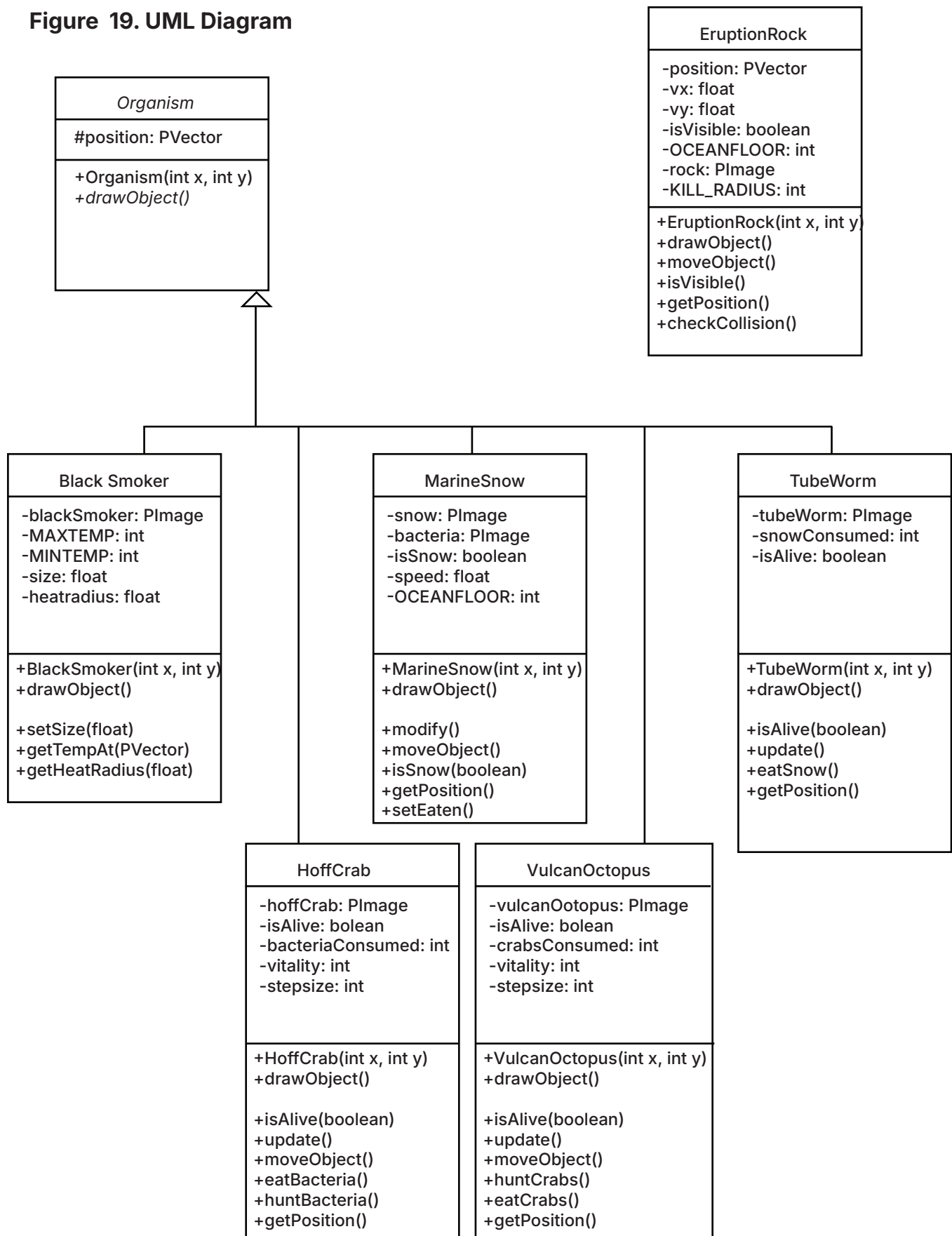
Figure 18. Thumbnail Sketch 3



Note. Environment Development

Appendix

Figure 19. UML Diagram



Note. UML Diagram for Extremophile farm